

AKASH BAG

Software Engineer | Frontend Engineer | Full-Stack Developer
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Professional Summary

Full-Stack Software Engineer with 2+ years of experience building scalable web applications, analytics dashboards, and data-driven systems using **React.js**, **Next.js**, **TypeScript**, **Node.js**, **PostgreSQL**, and **Docker**, with strong skills in API design, authentication, RBAC, and performance optimization.

Professional Experience

Software Engineer

Inwork Global Pvt Ltd | Nov 2025 – Jun 2026

- Developed **2+ analytics and operational dashboards** for KPI monitoring, data visualization, and real-time reporting, with **RBAC**, **protected routes**, and **permission-based UI rendering** across **4+ user roles**.
- Applied **Agentic AI workflows** and **AI-assisted development** to accelerate engineering delivery by **30%**.

EV Charging Portal Dashboard – Leading Indian Petroleum Company

- Contributed to the **overall architecture and data-flow models** of an EV charging portal dashboard across client, backend, and UI layers.
- Implemented **authentication and authorization with NextAuth**, including session handling, protected routes, and role-based access.
- Built a **bridge API layer** that calls the client's original APIs (backed by a separate database) and syncs data with our database, enabling communication between both systems.

Automotive Sales & Dealer Platform – Automotive Industry Client

- Architected a distributed query orchestration system handling **500K+ records** for analytical filtering, reporting, and CSV export workflows.
- Built asynchronous worker-based pipelines for long-running queries, achieving **sub-second API response times**.
- Optimized complex **SQL and PostgreSQL queries**, reducing analytical query execution time by **40%**.
- Implemented scheduled **cron jobs** to fetch data from an external **Salesforce** database, and designed a dedicated **salesman database** to store and serve the synced Salesforce data.
- Planned and executed a **database migration** with planned architecture, **indexing**, and **column mapping**.
- Designed a **VIN-structure database** to filter records by VIN number efficiently, significantly **reducing filtration time**.
- Built a secure **integration API** with authentication and default configuration settings to expose the platform inside a separate application, and worked with that team to debug integration issues.
- Delivered **major and minor UI/UX enhancements** across the application.

Software Development Engineer L-1

IEM-UEM Group, Kolkata | Sep 2024 – Oct 2025

- Led a team of **4 engineers** to architect and deliver a university hiring portal serving **1,000+ students and recruiters**, owning system architecture, task delegation, API integration, and technical decision-making.
- Engineered a real-time **industrial IoT monitoring platform** for machine health tracking, anomaly detection, predictive maintenance, and operational analytics across **50+ connected devices**.
- Built scalable **data ingestion pipelines** for high-frequency IoT telemetry with **sub-second refresh** across **4+ role-based dashboards**.
- Developed secure **REST API-driven workflows** for device management, telemetry processing, authentication, authorization, and role-based access control across operations and engineering teams.
- Built reusable frontend components and responsive dashboards using **React.js**, **JavaScript**, and **Tailwind CSS**.

Software Developer Intern

IEMA Research & Development Pvt Ltd | Mar 2024 – Aug 2024

- Optimized a university coding competition platform serving **12,000+ students** by integrating **REST APIs** and real-time leaderboard functionality, improving data refresh latency for live event scoring.
- Built reusable **React.js components**, API-integrated interfaces, and UI features used by **500+ active users**.

Front-End Developer Intern

ReadyToGo Travels Pvt Ltd | May 2022 – Jul 2022

- Developed responsive and user-focused frontend interfaces using **HTML5**, **CSS3**, and **JavaScript**, and participated in code reviews.

Technical Skills

Languages: JavaScript, TypeScript, C/C++, SQL, HTML5, CSS3

Frontend: React.js, Next.js, Tailwind CSS, React Query, Responsive Design, Component Architecture, State Management, API Integration

Backend: Node.js, Fastify.js, Express.js, Payload CMS, RESTful APIs, Authentication, Authorization, RBAC, Cron Jobs, Salesforce

Databases: PostgreSQL, MySQL, MongoDB, Firebase, SQL, Database Design, Database Migration, Indexing, Query Optimization

Tools & DevOps: Docker, Git, GitHub, CI/CD, Drizzle ORM, NextAuth

Architecture: Microservices, Distributed Systems, REST Architecture, RBAC, Scalable Systems, Event-Driven / Async Processing

Engineering Practices: Agile, Code Review, Debugging, Performance Optimization, API Design, AI-Assisted Development, Agentic AI

Education

B.Tech in Computer Science and Technology – University of Engineering & Management, Kolkata

CGPA: 8.8 / 10 | 2024

Achievements

- Contributed to open-source projects across multiple **Hacktoberfest** events, earning official recognition for community contributions.
- Achieved a maximum competitive programming rating of **1,704** and solved **200+ LeetCode problems** across multiple coding competitions.

Certifications

Complete React Developer (Redux, Hooks, GraphQL) – Udemy

2023

The Web Developer Bootcamp – Udemy

2023